Necessary face and contact conditions for a ramified selector

Use the canonical source order $c = (d, \lambda, n, m, C, A, D, \alpha, \beta) \geq 0$ and the matrices P, N, E, K of the canonical-completion note. Put $v = 16s^2$ and let $J(s) \in (\mathbb{R}[s]_{\leq 4})^{2\times 2}$. Write $R = K - E - J$ and

$$G = \begin{pmatrix} P+K & J \\ J^T & N+K \end{pmatrix}, \qquad H = \begin{pmatrix} P+E & R \\ R^T & N+E \end{pmatrix}.$$

The nine source coordinates remain independent. Positivity on $s \geq 0$ constrains the selector at both endpoints and at every zero of the physical state form.

Lemma 1 (Endpoint support and contact interpolation)**.** *Let $D_i^M(s)$ be diagonal entry i of $M = G, H$. With $\delta = d/2$, $D_i^M(s) = 4(a_i^M + 256 b_i^M s^4)$, where*

i	a_i^G	b_i^G	a_i^H	b_i^H
1	λ	m	A	$m + A + \alpha + \delta$
2	$m + \beta + n + \delta$	$\beta + \lambda + n + \delta$	$m + \beta + \alpha + \delta$	β
3	$C + D + \lambda + \delta$	C	$C + D + A + \delta$	$C + A + \alpha + \delta$
4	n	$D + \lambda + n + \delta$	α	D

For $i, j \in \{1, 2\}$, an entry Z_{ij}, with $Z = J$ for G and $Z = R$ for H, vanishes identically if either D_i^M or D_{j+2}^M does. Otherwise every exponent k occurring in it satisfies

$$\frac{\operatorname{ord}_0 D_i^M + \operatorname{ord}_0 D_{j+2}^M}{2} \leq k \leq \frac{\deg D_i^M + \deg D_{j+2}^M}{2}. \tag{1}$$

Here the orders and degrees of a nonzero diagonal are 0 or 4. Thus these conditions are explicit affine restrictions on the twenty coefficients of J, on every source face.

At an interior full-state contact $(U_0, s_0; a, b)$, where $U_0 > 0$, $s_0 > 0$, and the canonical form $\Phi_c(U_0, 16s_0^2; a, b) = 0$, put $g = (a, U_0 b)^T$, $h = (a, b)^T$. Then

$$G(s_0)g = 0, \qquad H(s_0)h = 0,$$

$$J(s_0)b = -(P+K)(s_0)a/U_0, \qquad J(s_0)^T a = -U_0(N+K)(s_0)b, \tag{2}$$

$$2U_0 a^T J'(s_0)b = -a^T(P+K)'(s_0)a - U_0^2 b^T(N+K)'(s_0)b. \tag{3}$$

Primes mean differentiation with respect to s. These formulas require no positive source coefficient and no nonzero state component.

Proof. The displayed diagonals follow by direct substitution; in particular the signed d terms cancel in the leading coefficients of H_{22} and H_{44}. All table entries are nonnegative. Every 2×2 principal minor gives $|Z_{ij}(s)|^2 \leq D_i^M(s) D_{j+2}^M(s)$. The lowest and highest powers as $s \downarrow 0$ and $s \to \infty$ prove (1), including all literal zero cases.

The exact identity

$$\Phi_c(U, 16s^2; a, b) = \begin{pmatrix} a \\ Ub \end{pmatrix}^T G(s) \begin{pmatrix} a \\ Ub \end{pmatrix} + U \begin{pmatrix} a \\ b \end{pmatrix}^T H(s) \begin{pmatrix} a \\ b \end{pmatrix}$$

makes both terms zero at a contact, so each PSD matrix kills its displayed vector. This gives (2). For a fixed kernel vector, the corresponding scalar quadratic form has a minimum at the interior point s_0; differentiating $g^T G(s) g$ gives (3). The analogous derivative from H is consistent: its sum with the G derivative, weighted by U_0, is $\partial_s \Phi_c = 0$. □

On C1, $\lambda = n = 0$, $d, m, C > 0$, the first support restriction gives

$$J(s) = \begin{pmatrix} p_2 s^2 + p_3 s^3 + p_4 s^4 & q_4 s^4 \\ r_0 + r_1 s + r_2 s^2 + r_3 s^3 + r_4 s^4 & \ell_2 s^2 + \ell_3 s^3 + \ell_4 s^4 \end{pmatrix}. \tag{4}$$

Only four coefficients are added to an unramified quadratic selector: p_3, ℓ_3, r_1, r_3. The second support restriction also retains the literal source faces. If $D = 0$, necessarily $J_{12} = 0$ and $J_{22} = \ell_2 s^2 + 512 d s^4$. If $\beta = 0$, necessarily $\deg J_{21} \le 2$ and $J_{22} = \ell_2 s^2 + 512 d s^4$. If both vanish, $J_{22} = 512 d s^4$. Additional zero sources are handled directly by the table; no passage to a strictly positive source is needed.

Positivity near a contact. The linear contact equations do not establish even local matrix positivity. For either $M = G, H$ at an interior contact, let V span $\ker M(s_0)$ and let T span a complementary subspace. Then $D = T^T M(s_0) T \succ 0$ and local positivity necessarily implies

$$V^T M'(s_0) V = 0, \qquad \frac{1}{2} V^T M''(s_0) V - (T^T M'(s_0) V)^T D^{-1} (T^T M'(s_0) V) \succeq 0. \tag{5}$$

This follows by expanding the Schur complement to second order. If the complementary space is empty, the inverse term is zero. At $s = 0$ only the one-sided condition $V^T M'(0) V \succeq 0$ is automatic; if that matrix vanishes, the second condition above applies. The same boundary rule holds at infinity after reversing the polynomial as $\tau^4 M(1/\tau)$.

The physical kernel compatibility makes the value equations exact: if $\mathcal{P}_c h = 0$ for the state matrix of Φ_c, then $Gg = 0$ implies $Hh = 0$ by its matrix splitting identity. When $a, b \neq 0$, the value equations impose three independent linear conditions. The difference of any two solutions is exactly $t\, a_\perp b_\perp^T$, where $a_\perp = (-a_2, a_1)^T$ and similarly for b. If $a = 0, b \neq 0$, the homogeneous freedom is $w b_\perp^T$; if $b = 0, a \neq 0$, it is $a_\perp w^T$, with $w \in \mathbb{R}^2$. The prescribed right sides must of course be compatible. The first jets impose scalar kernel expectations, not $G'g = 0$ or $H'h = 0$. The same twenty polynomial coefficients must satisfy these conditions at every contact, the two endpoint support restrictions, and the coupled Schur inequalities (5). Rank drops use the full kernel V, rather than only the physical contact vector. Even these second-order conditions are not sufficient: $-(s - s_0)^4$ passes the corresponding scalar zero, first- and second-derivative tests but is negative nearby.

Universal coverage requires

$$\forall c \in C_{\text{can}} \quad \exists J_0, \ldots, J_4 \in \mathbb{R}^{2\times 2} \quad \forall s \ge 0 : \quad G(s), H(s) \succeq 0.$$

The lemma reduces this question to a common quartic selector satisfying the endpoint and contact constraints while maintaining positivity between contacts. The existence of such a selector for every valid source remains open.